AQ1.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.

4.1 Level 1:

Let AA be a $3 \times 3$3 × 3 matrix with non-zero determinant. If $det(2A) = k\ det(A)$det(2A) = k det(A), then what will be the value of kk?

[Hint: If a scalar (c)(c) is multiplied with one row of a matrix AA, then the determinant of the new matrix will be cc times the determinant of AA.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

1 point

1 point

If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \end{bmatrix}$A = [2−11−102] and $B = \begin{bmatrix} 1 & 3 & -2 \\ 0 & -1 & 1 \\ 3 & 4 & 2 \end{bmatrix}$B = [103−1342−21], then choose the set of correct options.

- $det(A) = -9$ and $det(B) = -3$det(A) = −9 and det(B) = −3.
- $det(A) = 9$ and $det(B) = 3$det(A) = 9 and det(B) = 3.
- $det(A) = -9$ and $det(B) = 3$det(A) = −9 and det(B) = 3.
- $det(AB) = -27$ and $det(BA) = 27$det(AB) = −27 and det(BA) = 27.
- $det(AB) = det(BA) = -27$det(AB) = det(BA) = −27.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A) = -9$ and $det(B) = 3$det(A) = −9 and det(B) = 3.

$det(AB) = det(BA) = -27$det(AB) = det(BA) = −27.

1 point

Let AA be a $2 \times 2$2 × 2 matrix, which is given as $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$[a11a21a12a22]. Define the following matrices :

$B = \begin{bmatrix} a_{11} & -a_{21} \\ a_{12} & -a_{22} \end{bmatrix}$B = [a11 − a21a21a12 − a22a22], $C = \begin{bmatrix} a_{11} & -a_{12} \\ a_{21} & -a_{22} \end{bmatrix}$C = [a11 − a12a21 − a22a12a22],

$D = \begin{bmatrix} a_{11} & +a_{21} \\ a_{12} & -a_{22} \end{bmatrix}$D = [a11 + a21a21a12 − a22a22], $E = \begin{bmatrix} a_{11} & -a_{21} \\ a_{12} & +a_{22} \end{bmatrix}$E = [a11 − a21a21a12 + a22a22]

Which of the matrices among $B, C, D,$ and EB, C, D, and E have the same determinant as that of the matrix AA, for any real numbers $a_{11}, a_{12}, a_{21}, a_{22}$ a11, a12, a21, a22?

- BB and DD
- BB and EE
- BB and CC
- DD and EE
- CC and EE

No, the answer is incorrect.
Score: 0
Accepted Answers:
BB and CC

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = [a11ta11 − sa31ra31a12ta12 − sa32ra32a13ta13 − sa33ra33] be a matrix and $r, s, t \neq 0$r, s, t = 0. Find $det(A)$det(A)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Suppose $A = \begin{bmatrix} -1 & 4 & -2 \\ 0 & 1 & 3 \\ 1 & 0 & -2 \end{bmatrix}$A = [−101410−23−2] and $B = \begin{bmatrix} 2 & 0 & -1 \\ 4 & \alpha & 0 \\ 0 & -3 & -5 \end{bmatrix}$B = [2400α−3−10−5]. If $det(AB) = 32$det(AB) = 32, then find the value of αα.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

4.2 Level 2:

(Use this information to answer questions 6 and 7)

Let $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$A(x1, y1), B(x2, y2) and C(x3, y3) be the vertices of a

triangle. The area of the triangle ABCABC is given by $\frac{1}{2}|det(D)|$21|det(D)| square units, where

$D = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$D = [x1x2x3y1y2y3111]

If $P(-1, 1), Q(1, 1)$ and $R(1, 0)$P(−1, 1), Q(1, 1) and R(1, 0) are the vertices of a triangle, then find the area (in square units) of the triangle PQRPQR.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
Let $P(x - 1, 2x), Q(0, x - 2)$ and $R(-x + 1, 1)$P(x − 1, 2x), Q(0, x − 2) and R(−x + 1, 1) be the vertices of a triangle. If the area of the triangle is 55 square units and $x > 0$x > 0, then choose the set of correct options

- [ ] length of side $PQ = \sqrt{29}$PQ = $\sqrt{29}$ units.
- [ ] length of side $PR = \sqrt{40}$PR = $\sqrt{40}$ units.
- [ ] length of side QR = 2QR = 2 units.
- [ ] length of side QR = 1QR = 1 units.

No, the answer is incorrect.
Score: 0
Accepted Answers:
length of side $PQ = \sqrt{29}$PQ = $\sqrt{29}$ units.
length of side QR = 2QR = 2 units.
1 point

Find out the correct value of $det(A)$det(A), where $A = \begin{bmatrix} 120102020 \times 2030 \\ 120202030 \times 2010 \\ 120302010 \times 2020 \end{bmatrix}$A = [111201020202030 2020 × 20302030 × 20102010 × 2020]

[[Hint: Observe that the given matrix is of the form: $\begin{bmatrix} 1 & a & b & c \\ 1 & b & c & a \\ 1 & c & a & b \end{bmatrix}$ [111abcbccaab]]]

○
00
○
200200
○
−2000−2000
○
20002000

No, the answer is incorrect.
Score: 0
Accepted Answers:
20002000

1 point

Let AA be a square matrix of order 33 and BB be a matrix that is obtained by adding 22 times the first row of AA to the third row of AA and adding 33 times the second row of AA to the first row of AA. What is the value of $det(6A^2 B^{-1})$det(6A2B−1)?

○
6 $det(A)$6 det(A)
○
6 $det(A)det(B)$6 det(A)det(B)
○
6^3 $det(A)$63 det(A)
○
6^3 $det(A)det(B)$63 det(A)det(B)

No, the answer is incorrect.
Score: 0
Accepted Answers:
6^3 $det(A)$63 det(A)

Suppose A = $\begin{bmatrix} 192021 \\ 202119 \\ 211920 \end{bmatrix}$A = $\begin{bmatrix} 192021 & 202119 & 211920 \end{bmatrix}$. Find the value of $\frac{1}{60} det(A^2)$601det(A2).

[[You can try to compute the determinant of a matrix of the form: $\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ $\begin{bmatrix} a & b & c & b & c & a & c & a & b \end{bmatrix}$]]

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 540

1 point